

University English Teaching in the Artificial Intelligence Era

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Abstract

The rapid advancement of artificial intelligence (AI) technologies—such as natural language processing (NLP), machine learning (ML), and adaptive learning systems—has revolutionized global educational paradigms, exerting profound and multifaceted implications for university English teaching, especially in the context of English as a Foreign Language (EFL) environments. This paper systematically explores the transformative impact of AI on English language education, with a focused analysis of its practical applications across four core dimensions: personalized learning (tailored to students' proficiency levels and learning paces), intelligent assessment (automated and real-time evaluation of language skills), interactive instruction (immersive tools like virtual reality and AI chatbots), and cross-cultural communication (simulating authentic intercultural scenarios). By examining current classroom practices, existing implementation challenges, and emerging development trends, this study argues that AI-driven tools can significantly enhance teaching efficiency (e.g., reducing teachers' routine workload), foster sustained student engagement (via gamified elements and instant feedback), and promote learner autonomy (enabling self-directed skill improvement). However, these benefits can only be fully realized if educators proactively adapt their pedagogical strategies to integrate AI in purposeful, education-centric ways. Critical issues must be addressed to maximize AI's value, including students' over-reliance on technology (which may hinder critical thinking and real-world communication skills), ethical concerns (such as student data privacy and algorithmic bias), and the digital divide (disparities in access to AI tools among students in different regions or with disabilities). Ultimately, this study emphasizes that AI should be viewed as a complementary tool that empowers both educators and learners—supporting teaching innovation and personalized learning—rather than a replacement for human interaction, which remains essential for fostering emotional connection, pragmatic language use, and cultural awareness in language education.

Keywords

Artificial Intelligence; University English Teaching; Personalized Learning; Intelligent Assessment; Pedagogical Innovation

1. Introduction

The 21st century has witnessed an unprecedented technological revolution driven by artificial intelligence (AI), which has permeated nearly every sector, including education. In university English teaching, AI technologies such as natural language processing (NLP), machine learning (ML), and adaptive learning systems are reshaping classroom

dynamics, offering new opportunities for innovation. According to a report by UNESCO (2021), AI in education has the potential to address global challenges such as educational inequality and teacher shortages by providing scalable, personalized learning experiences. In the context of English as a Foreign Language (EFL) teaching, AI tools can simulate real-life communication scenarios, provide instant feedback, and tailor instruction to individual student needs, thereby bridging gaps in traditional teaching methods.

However, the integration of AI into university English curricula is not without controversy. Critics argue that excessive dependence on AI may hinder the development of critical thinking and interpersonal skills, which are essential for language acquisition. This paper aims to examine the multifaceted role of AI in university English teaching, exploring its practical applications, benefits, challenges, and future directions. By synthesizing existing research and case studies, this study seeks to provide insights for educators, policymakers, and technologists striving to harness the potential of AI to enhance language education.

2. The Theoretical Foundation of AI in Language Teaching

To understand AI's impact on university English teaching, it is essential to ground the discussion in relevant educational theories. Constructivism—a learning theory emphasizing active knowledge construction through interaction and reflection, aligns with the ability of AI to create immersive, student-centered environments (Piaget, 1970). AI-driven platforms enable learners to engage with dynamic content, collaborate with peers, and receive personalized guidance, fostering deeper understanding and retention.

Another key framework is sociocultural theory, which highlights the role of social interaction in language development (Vygotsky, 1978). AI tools such as chatbots and virtual reality (VR) simulations facilitate authentic communication by providing learners with opportunities to interact with “native speakers” or participate in real-world scenarios (e.g., job interviews, academic conferences) in a low-pressure environment. This aligns with the communicative language teaching (CLT) approach, which prioritizes practical language use over rote memorization.

Additionally, adaptive learning theory, which posits that instruction should be tailored to individuals' learning styles, pace, and abilities, is central to the design of AI. Machine learning algorithms analyze student performance data to identify strengths and weaknesses, adjusting content difficulty and feedback accordingly (Brusilovsky, 2019). This personalized approach addresses the diverse needs of university students, who often have varying proficiency levels, learning goals, and cultural backgrounds.

3. Practical Applications of AI in University English Teaching

AI's applications in university English teaching are diverse and rapidly expanding. This section examines four key areas: personalized learning, intelligent assessment, interactive instruction, and cross-cultural communication. For example, artificial intelligence (AI) has revolutionized university English teaching by offering personalized and interactive learning experiences. AI-powered language learning platforms such as Grammarly and Pigaiwang provide real-time feedback on students' writing, identifying grammatical errors, improving sentence structure, and suggesting vocabulary enhancements. This instant feedback allows students to revise their work independently and develop writing skills more efficiently.

Intelligent tutoring systems like Duolingo and Babbel use adaptive learning algorithms to create personalized study plans based on students' proficiency levels and learning pace. These platforms offer targeted exercises in listening, speaking, reading, and writing, ensuring that each student receives customized practice opportunities.

Virtual language assistants, powered by natural language processing technology, enable students to practice conversational English anytime, anywhere. Tools like ELSA Speak provide pronunciation assessment and feedback, helping learners improve their speaking skills through AI-driven voice recognition.

AI also facilitates automated grading of objective assessments, saving teachers valuable time and allowing them to focus on providing quality feedback on subjective assignments. Additionally, learning analytics tools help educators identify students' learning patterns and difficulties, enabling timely interventions and curriculum adjustments.

3.1 Personalized Learning

One of AI's most significant contributions to English teaching is its ability to deliver personalized learning experiences. Adaptive learning platforms, such as Coursera's “AI Tutor” and Duolingo's “Personalized Practice,” use ML algorithms to analyze student data (Xu, 2023) (e.g., test scores, response times, error patterns) and generate

customized learning paths. For example, a student struggling with grammar might receive targeted exercises and video tutorials, while an advanced learner could be challenged with complex reading comprehension tasks or academic writing assignments.

In university settings, AI-powered learning management systems (LMS) like Blackboard and Canvas integrate with tools such as Grammarly and Turnitin to provide real-time feedback on writing. Grammarly's NLP technology not only corrects grammatical errors but also suggests improvements in clarity, tone, and style, helping students refine their academic writing skills. Similarly, AI-driven reading platforms like LingQ analyze texts to identify vocabulary gaps, allowing learners to create personalized word lists and review materials based on their progress.

Take Xuzhou University of Technology as an example: Freshmen engage in independent college English learning using the We-Learn English learning platform developed by Foreign Language Teaching and Research Press. This platform records students' learning situations, including article reading, after-class exercises, in-class tests, and extended training. It effectively helps freshmen quickly adapt to the rhythm of college English learning. At the end of the semester, it will summarize students' learning situations, including learning duration and test completion, and give scores. This helps teachers aggregate students' routine academic performance, greatly improving teaching efficiency and enhancing teaching effectiveness.

3.2 Intelligent Assessment

Traditional assessment methods, such as exams and essays, are often time-consuming to grade and may not capture students' holistic language abilities. AI-driven assessment tools address these limitations by offering efficient, objective, and multifaceted evaluation.

Automated essay scoring (AES) systems (Li, 2024), such as ETS's e-rater and Pearson's Intelligent Essay Assessor, use NLP to evaluate writing quality based on criteria like grammar, coherence, vocabulary, and argumentation. These tools provide instant feedback, allowing students to revise their work promptly and instructors to focus on higher-order skills (e.g., critical thinking, creativity). Research by Shermis and Burstein (2013) found that AES scores are comparable to human raters in terms of reliability, making them a viable supplement to traditional grading.

AI also enables formative assessment through continuous data collection. For instance, language learning apps like Babbel track students' daily practice, identifying patterns in their mistakes (e.g., confusion between present perfect and simple past tense) and recommending targeted exercises. This real-time feedback helps students address weaknesses before they become ingrained, enhancing learning efficiency.

3.3 Interactive Instruction

Artificial intelligence has transformed classroom teaching by introducing interactive tools that can engage students and boost their engagement. In college English teaching, interactive teaching (Praktika, 2025) utilizes technologies such as AI dialogue systems and virtual scenario drills to create an immersive language environment. Students can complete real-time oral assessments through voice interaction, engage in multiple rounds of conversations with AI-powered virtual partners during role-playing tasks, and receive real-time feedback on grammar errors and suggestions for pronunciation optimization from the system. The intelligent question bank dynamically generates personalized exercises based on students' levels, while the classroom answerer and group collaboration platform break the traditional teaching mode, allowing teacher-student interaction and student-student interaction to run through the entire process of listening, speaking, reading, and writing. This student-centered interactive teaching not only enhances the fun of learning, but also effectively improves language application ability and deep learning effectiveness through high-frequency feedback and collaborative tasks.

VR and augmented reality (AR) technologies further enhance interactivity by creating immersive learning environments. For example, the VR platform "Language Lab VR" allows students to explore virtual cities like London or New York, interacting with locals, ordering food in restaurants, and navigating public transportation—all while practicing English in context. Such experiences not only make learning more engaging but also improve knowledge retention by linking language to vivid, multisensory memories (Mayer, 2009).

Another innovative application is speech recognition technology, which helps students improve pronunciation and fluency. Tools like ELSA Speak use AI to analyze students' speech, comparing it to native speaker models and providing feedback on intonation, stress, and articulation. This is particularly valuable for university students preparing for oral exams (e.g., IELTS, TOEFL) or future careers requiring clear communication.

3.4 Cross-cultural Communication

Cross-cultural communication is a crucial part of college English teaching, emphasizing that teaching should not only focus on imparting language skills, but also on cultivating students' cultural understanding and communication abilities. For example, when teaching of an article about the Sino-U.S. educational differences in the new edition of *Advanced English: Volume 3*, instructors can compare the two countries' educational methods. American education respects children's nature and allows them to develop freely; Chinese education, on the other hand, tends to be more hands-on-for instance, helping children unlock locks instead of letting them try to do so themselves. Teachers can use such cases to guide students to perceive the role of cultural context in language expression. In addition, the video materials in the textbook compare the differences between Chinese and English education, allowing students to understand the differences between Chinese and English education through real cases and cultivate critical thinking.

4. Benefits of AI in University English Teaching

The integration of AI into university English teaching offers numerous benefits for both students and educators.

4.1 Enhanced Learning Efficiency

Artificial intelligence can provide personalized teaching and instant feedback, allowing students to focus on areas that need improvement, thereby accelerating learning. For example, the language learning platform We-Learn helps students preview lessons in advance, review content after class, learn vocabulary, understand difficult sentences, complete homework, and take-in class assessments. For busy college students who balance multiple responsibilities, it is a highly efficient way of learning.

4.2 Increased Student Engagement

Artificial intelligence driven tools make learning more interactive and fun, solving the problem of student apathy in large lecture halls. Gamified elements such as badges, leaderboards, and rewards motivate students to actively participate. For example, the language learning application We-Learn prompts students to check in manually before class, administers in class tests during class, and keeps track of academic task completion, reminding them of their English learning progress.

4.3 Improved Accessibility

AI helps bridge the digital divide by providing flexible learning options for students with disabilities or those in remote areas. For instance, text-to-speech tools assist visually impaired students, while captioning and translation features support non-native speakers. AI-powered platforms also enable asynchronous learning, allowing students to access materials and complete assignments at their own pace, regardless of time or location.

4.4 Data-driven Decision Making

AI generates vast amounts of data on student performance, enabling educators to identify trends, evaluate teaching effectiveness, and adjust curricula accordingly. For example, if data reveals that most students struggle with academic writing, instructors can design targeted workshops or update course materials. This evidence-based approach ensures that teaching strategies are aligned with student needs, improving overall learning outcomes.

5. Challenges and Ethical Considerations

Despite its benefits, AI integration in university English teaching faces significant challenges and ethical dilemmas.

5.1 Over-reliance on Technology

Overreliance on artificial intelligence may weaken students' critical thinking and effective communication skills in real-world environments. For example, during class, when an instructor assigns a composition, students may take out their phones and use AI tools such as DeepSeek, Doubao, and Kimi to help them write. This causes students to not think and rely on AI to help write the composition. At this time, the teacher should guide students in a timely manner, telling them to think first before writing, and feed their own compositions to AI for help modify and correct. This,

however, requires students to have stronger English expression skills. Teachers must strike a balance between technology and interpersonal interaction, using artificial intelligence as a supplement to traditional teaching rather than a substitute.

5.2 Privacy and Data Security

AI platforms collect sensitive student data, including personal information, learning habits, and performance records. This raises concerns about data privacy and the risk of unauthorized access or misuse. Universities must implement robust security measures and transparent data policies to protect student confidentiality.

5.3 Bias and Cultural Insensitivity

Artificial intelligence technology brings not only opportunities for efficiency improvement in college English teaching, but also challenges in cultural education. Its bias and cultural insensitivity essentially reflect the social and cultural power structure behind technology, which permeates into the teaching process through educational tools and may have a profound impact on students' language and cultural perspectives. As implementers of cross-cultural education, the core responsibility of college English teachers is not to resist technological progress, but to establish a cultural consciousness of technological application - to not only make good use of AI tools to improve teaching efficiency, but also to keenly identify cultural deviations in technological output, and guide students to form comprehensive language cognition and open cultural attitudes through critical teaching practices.

In this regard, college English teachers need to pay attention to three aspects: first, they need to use AI tools critically, actively identify and correct biases in the system outputs, such as pointing out stereotypes in AI's description of "female professions"; The second is to balance AI resources with cultural teaching objectives, supplement local cultural English materials, and cultivate students' two-way cross-cultural communication skills; Thirdly, AI evaluation results cannot be used as the sole criterion. It is necessary to comprehensively evaluate student performance based on dimensions such as pragmatic appropriateness and cultural adaptability, in order to avoid narrowing teaching objectives with technological tools.

6. Future Directions and Recommendations

To maximize AI's potential in university English teaching, stakeholders must collaborate to address challenges and embrace innovation.

6.1 Integrating AI with Human Expertise

The future of language education lies in a hybrid model that combines AI's efficiency with human teachers' empathy and expertise. Educators should focus on roles AI cannot replace, such as mentoring, emotional support, and fostering critical thinking. For example, AI can handle routine tasks like grading and vocabulary drills, freeing instructors to lead discussions, provide personalized feedback on complex writing assignments, and guide students' career development.

6.2 Developing Ethical AI Guidelines

Universities and tech companies must work together to establish ethical standards for AI use in education. This includes ensuring transparency in algorithm design, protecting student data, and mitigating bias. Educators should also teach students digital literacy skills, equipping them evaluate AI-generated content critically and use technology responsibly.

6.3 Investing in Teacher Training

To effectively integrate AI into the classroom, educators need training in both technological skills and pedagogical innovation. Professional development programs should focus on AI tool selection, data analysis, and curriculum design, empowering educators to make informed decisions about technology integration. Partnerships between universities and tech companies can provide workshops, certifications, and ongoing support for educators.

7. Conclusion

Artificial intelligence is transforming university English teaching by offering personalized, interactive, and efficient learning experiences. From adaptive platforms to VR simulations, AI tools align with modern educational theories and address the diverse needs of today's students. However, realizing AI's full potential requires addressing challenges such as over-reliance on technology, privacy concerns, and ethical biases. By integrating AI with human expertise, investing in educator training, and promoting inclusive development, stakeholders can harness this technology to create a more engaging, accessible, and effective language education system.

As AI continues to evolve, its role in university English teaching will only grow. Educators must embrace this change as an opportunity to reimagine pedagogy, empowering students to become confident, competent, and culturally aware communicators in the globalized world. Ultimately, AI is not a replacement for human teachers but a powerful tool that, when used thoughtfully, can enhance the teaching and learning process, preparing students for success in the 21st century and beyond.

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